# MITS 5300G – Assignment 2

**Instructions:**

- **Submit your answers through Canvas, as a single zip file. Submitting this file (and/or others as indicated within the assignment) in alternate formats or naming conventions will result in a 1 point penalty. It is your responsibility to verify that the assignment was properly submitted through Canvas by the deadline.**
- **Please ensure you are familiar with the course policy on academic integrity. This assignment is to be completed and written up individually.**
- **Watch your spelling and grammar; you will lose marks for spelling and grammar mistakes.**
- **Keep all answers short, clear, and concise. Writing more sentences than necessary will not result in a better mark.**

**Background:** Before proceeding with this assignment, it is strongly recommended that you go through the SELinuxPolicyTutorialFiles page, and its links (under the SELinux Part 2 module) on Canvas, to confine the hello world example program.

**Goal:** Using the Fedora Security Labs VM you created for Assignment 1, write your own SELinux policy module for the provided target application (the source for this application is attached to the assignment as a ".c" file; read the source code comments for compilation instructions). The compiled target application is to be run from a new user account called **Adm_<FirstnameLastname>**, where *<Firstname> is your own first name and <Lastname> your own last name*), and placed in this user's home directory. Name the compiled target application <***FirstnameLastname>_app***, You must also apply your policy such that it ensures that:

- The <***FirstnameLastname>_app*** application will be confined within a new domain called <FirstnameLastname>_t .
- When the <***FirstnameLastname>_app*** is confined in <FirstnameLastname>_t, that it:
  - remains fully functional,
  - has *minimum permissions* to *only* the files it requires to run properly, and
  - runs with no error messages.
  - **Do not modify any of the SELinux booleans during your policy creation process.**

**Additional Notes:**

- You will need to use the "audit2allow" program when trying to determine some of the (unexpected) policy rules required. The process will likely require multiple iterations.

1. (5 points) Please submit the answer to this question in a file named <***YourFullName>_commands.pdf, included in your .zip file submission***. All screenshots must include the prompt showing it is run from your Adm_<FirstnameLastname> account.

    a. (2.5 points) Show a screenshot of the compiled target application running in your own domain <FirstnameLastname>_t. Hint: after a certain point, you could even just have the domain in permissive mode to obtain this screenshot, but note that will need to change to complete the rest of this assignment.

    b. (1.5 points) Submit a screenshot of the additional commands (entered on the Linux command line) needed to enter in order to permanently set up the target application so that it is confined, and to support its *minimum permissions* to *only* those files it requires to run properly. Hint: remember to consider what label(s), if any, you need to give the following files: the compiled target application, the params file, and the records file. Also, consider what steps were needed prior to any labeling.

    c. (1 point) Submit a screenshot of the additional commands (entered on the Linux command line), along with their output, that *you used to verify* that the target application is running confined in the <FirstnameLastname>_t domain and that all of the relevant files are labelled with the correct types.

2. (3 points) Make sure that your policy keeps the necessary rules only, and does not contain any extra (unneeded) rules. Explain why the rules you kept are necessary. Provide these details in a comment in your policy file, named <***FirstnameLastname> _myFirstPolicy.te,*** for each line explaining what the following line in the policy does and why it is needed. Include your .te file in your .zip file submission.